

Tarek Mohamed Salah

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Summary

Front-End Developer and senior Electronics and Communication Engineering student with strong skills in HTML, CSS, JavaScript, and React. Experienced in building responsive, user-friendly interfaces and applying OOP principles from a background in C and Python. Familiar with Git, API integration, and modern development tools. Eager to contribute to front-end projects and grow in a collaborative tech environment.

Education

Alexandria University

BS in Communication and Electronics Engineering

GPA: 3.41/4.0, Excellent

Alexandria, Egypt

Sept 2021 – Sept 2026

- Summary: Gained in-depth knowledge and practical experience in analog and digital circuit design, signal processing, transmission systems, and embedded systems. Developed strong technical skills through coursework and projects involving MATLAB, Simulink, Embedded C, Assembly Language, LaTeX/Overleaf, and operating systems. Also experienced in project management and data structures & algorithms.
- Core Coursework & Tools: Digital Communication · Analog & Digital Circuit Design · Embedded Systems · Optical Transmission · Digital Signal Processing · Operating Systems · Data Structures & Algorithms · MATLAB · Simulink · Assembly Language · Embedded C · Python · C · LaTeX / Overleaf · Project Management

Experience

Software Engineer Intern

Alex Eagles Aero Design

Alexandria, Egypt

Sept 2024 – present

- Developed Python scripts to enable autonomous drone movement and obstacle avoidance.
- Implemented mapping techniques for terrain analysis and real-time decision-making.
- Contributed to problem-solving strategies for enhancing drone autonomy and efficiency.

Software Engineer Intern

ALX

Remotely

Feb 2023 – Feb 2024

- Summary:
Completed intensive, hands-on training in software engineering fundamentals through real-world projects and peer collaboration. Developed strong skills in system programming, data structures, and backend development with a DevOps mindset.
- Key Skills & Tools:
C Programming · Python · Data Structures & Algorithms Linux · Unix Programming · Operating Systems
Git & GitHub · Shell Scripting · Reverse Engineering (Intro) Infrastructure Design & Management (Intro)
Back-End Development (Basics) Problem Solving · Debugging · Time Management · Collaboration

Projects

Autonomous Drone System

Sept 2024 – June 2025

A whole system for Autonomous Drone

- Developed and tested the drone control class using DroneKit-Python and MAVLink protocol.
- Participated in backend–frontend integration for real-time mission updates and map visualization.
- Contributed to modules like lawn_pattern, drone-class-using-dronekit, and mission planning features.

- Collaborated on a multi-branch GitHub workflow using Git, Python, JavaScript, and React.
- Worked in a team of 6 developers focusing on modules such as global path planning, SDK integration, UI/UX, and FTS settings.
- Applied knowledge of embedded systems, aerodynamics, and software architecture in UAV control.

Skills

Front-End Development: HTML5, CSS3, JavaScript (ES6+), React.js

Responsive Design (Flexbox, CSS Grid, Media Queries)

TypeScript (in progress), UI/UX Principles

Git & GitHub, Chrome DevTools, VS Code

API Integration, Component-Based Architecture

Programming Languages: C, Python, JavaScript, Embedded C, Assembly Language

Familiar with Java, Shell Scripting (Unix), Python Scripts

Tools: Git & GitHub VS Code, Chrome DevTools MATLAB Simulink LaTeX / Overleaf Multisim

Languages: English (Expert), Arabic (native)

Soft Skills: Team Collaboration, Leadership & Communication, Customer Service (Call Center Experience), Project Management, Problem Solving and Infrastructure Design Awareness